

Skills Summary

Exponential Functions: Evaluating, Graphing, Growth and Decay

Use this reference while completing your worksheet or when studying.

Every function on your worksheet has the same shape, $f(x) = a \cdot b^x$: a starting value a multiplied by a base b once for each step of x . Read a as “where it starts” and b as “what it does each step”.

1. Evaluating an exponential function

Rule: In $f(x) = a \cdot b^x$, do the *power first*, then multiply by a . $a \cdot b^x$ never means $(a \cdot b)^x$.

Two facts that cover the awkward exponents: $b^0 = 1$ for any nonzero base, and $b^{-n} = \frac{1}{b^n}$ — a negative exponent gives a small positive answer, never a negative one.

Example: For $f(x) = 4 \cdot 3^x$, find $f(2)$.

Step 1. The input replaces x everywhere, and the order of operations puts the power ahead of the multiplication.

Step 2. $3^2 = 9$, and then $4 \cdot 9$ finishes it.

$$f(2) = 36$$

Try it: For $g(x) = 5 \cdot 2^x$, find $g(3)$.

check: 40

2. Graphing an exponential function and reading it

How to graph: build a short table (use $x = -1, 0, 1, 2, 3$), plot the points, and join them with a *smooth curve* — never a straight line.

What the picture always shows: the curve crosses the y -axis at a ; it rises to the right when $b > 1$ (growth) and falls to the right when $0 < b < 1$ (decay); it approaches the x -axis but never touches it.

Reading backwards: to find the x that gives a chosen height, write both sides as powers of the same base and match the exponents.

Example: Graph $f(x) = 2^x$ and find the x where the curve reaches height 8.

Step 1. The table gives the points to plot, and the same table answers the backwards question, so build it first.

Step 2. $f(0) = 1$, $f(1) = 2$, $f(2) = 4$, $f(3) = 8$: the height 8 is reached at the fourth point, and $2^x = 8 = 2^3$ confirms the exponents match.

$$x = 3$$

Try it: On the graph of $f(x) = 3^x$, at what x is the height 81?

check: 4 = x

3. Growth and decay models from a percent

Rule: A percent change becomes a multiplier, applied once per period. Growing by r percent gives base $b = 1 + \frac{r}{100}$; shrinking by r percent gives $b = 1 - \frac{r}{100}$.

So 6 percent growth is $b = 1.06$, and 12 percent decay is $b = 0.88$ — the amount *kept*, not the amount lost.

Example: A car worth 18000 dollars loses 12 percent of its value each year. What is it worth after 3 years?

Step 1. Losing a fixed percent each year is repeated multiplication, so this is exponential decay and the base is the fraction kept.

Step 2. Kept each year: $1 - 0.12 = 0.88$, so the value is $18000 \cdot (0.88)^3 = 18000 \cdot 0.681472$.
12266.50 dollars

Try it: A town of 5000 people grows 4 percent per year. Find the population after 6 years, to the nearest whole person.

2729 people

(!) Watch out: $3 \cdot 2^4$ is 48, not 6^4 — the coefficient is not part of the base. And a decay percent is not the base: losing 15 percent means multiplying by 0.85 every year, not by 0.15.